



Metascience, replicability, and META-REP

Guest lecture & workshop

Moritz Schiltenswolf

LMU Munich



„The use of scientific methods on the scientific process itself“
„Research on research“

Metascience includes a range of topics, such as:

- ➔ Open science.
- ➔ Beliefs and standards of researchers.
- ➔ Publication system, peer review.
- ➔ Incentives in the academic system.
- ➔ Replicability/Reproducibility of research.



ELSEVIER

SPRINGER NATURE

WILEY

Taylor & Francis

SAGE Publishing



frontiers

PLOS

MDPI



CAMBRIDGE UNIVERSITY PRESS

- Science aims to discover regularities.
- Scientific studies seek to provide evidence for theoretical claims that describe such regularities.
 - For example, a statistically significant difference between experimental conditions may support a specific theoretical prediction.
- If these regularities are genuine, repeated tests should produce consistent results.
- Consequently, two studies testing the same claim with equally valid methods should arrive at similar conclusions.
 - Replication studies

Replicability in psychology.

First studies indicating problems

Case 1 – Behavioral Priming

- In 1996, Bargh et al. published a study indicating that participants walked slower after being primed with an „Old“ stereotype.



- Although widely known and taught, the effect could not be replicated. (Doyen et al., 2012)

Case 2 – Precognition

- Bem (2011) reported evidence of „psi“, that is precognition about future events.



- The results could not be reproduced using the same data. (Wagenmakers et al. 2011)
- The results could not be replicated after collecting new data. (Galak et al., 2011)

Many Labs Project

- Growing concern about study replicability in psychological research. However, there was no tradition of replicating the work of others'.
- To change this Brian Nosek, Richard Klein, Olivia Atherton, and Charles Ebersole initiated the *Many Labs Project*
- Over the coming years, they initiated multiple replication projects in which researchers from many labs replicated a selection of studies.

RESEARCH ARTICLE SUMMARY

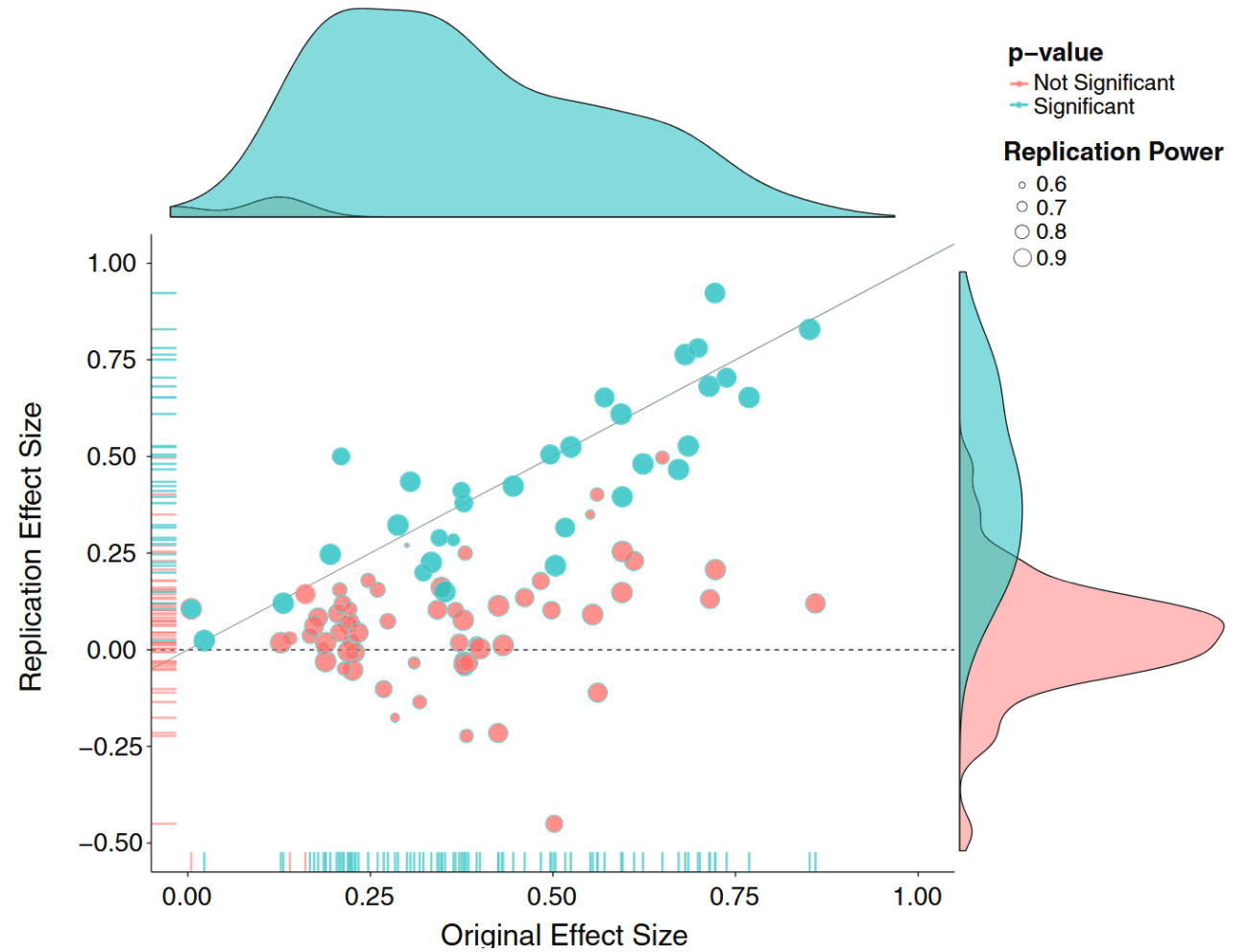
PSYCHOLOGY

Estimating the reproducibility of psychological science

Open Science Collaboration*

„The spark“ (Open Science Collaboration, 2015)

Estimating the reproducibility of psychological science

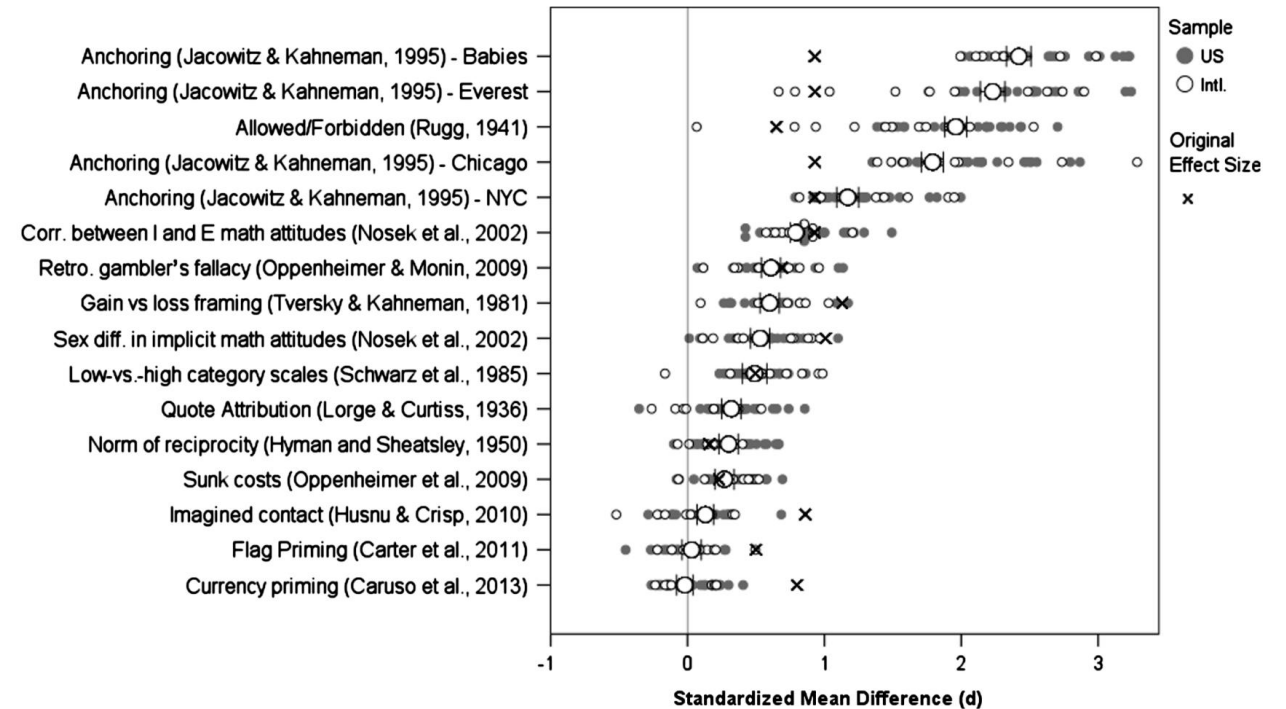


Many Labs Project

„The spark“ (Open Science Collaboration, 2015)

Many Labs 1 (Klein et al., 2014)

Investigating Variation in Replicability



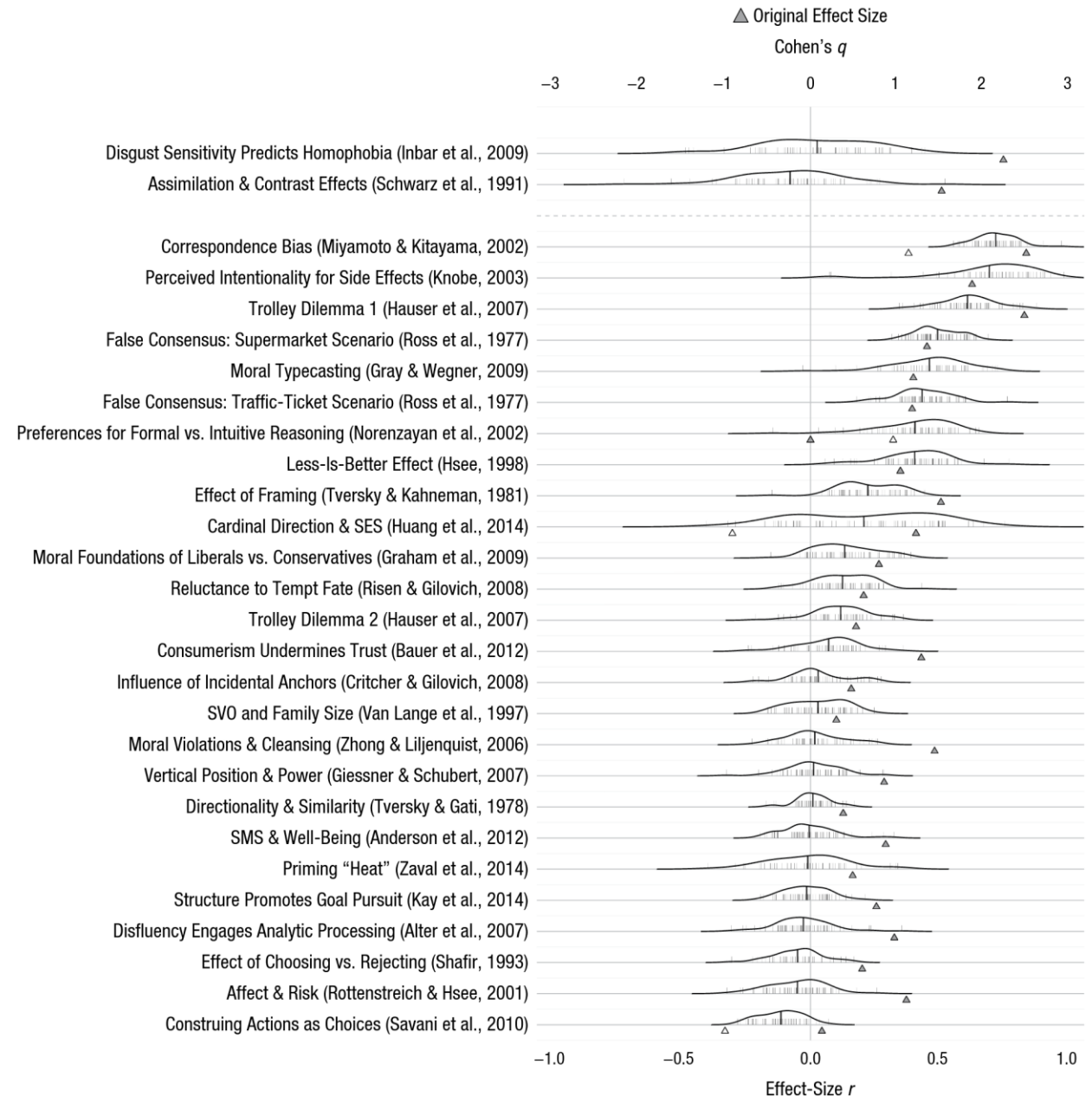
Many Labs Project

„The spark“ (Open Science Collaboration, 2015)

Many Labs 1 (Klein et al., 2014)

Many Labs 2 (Klein et al., 2018)

*Investigating Variation in Replicability
Across Samples and Settings*



Many Labs Project

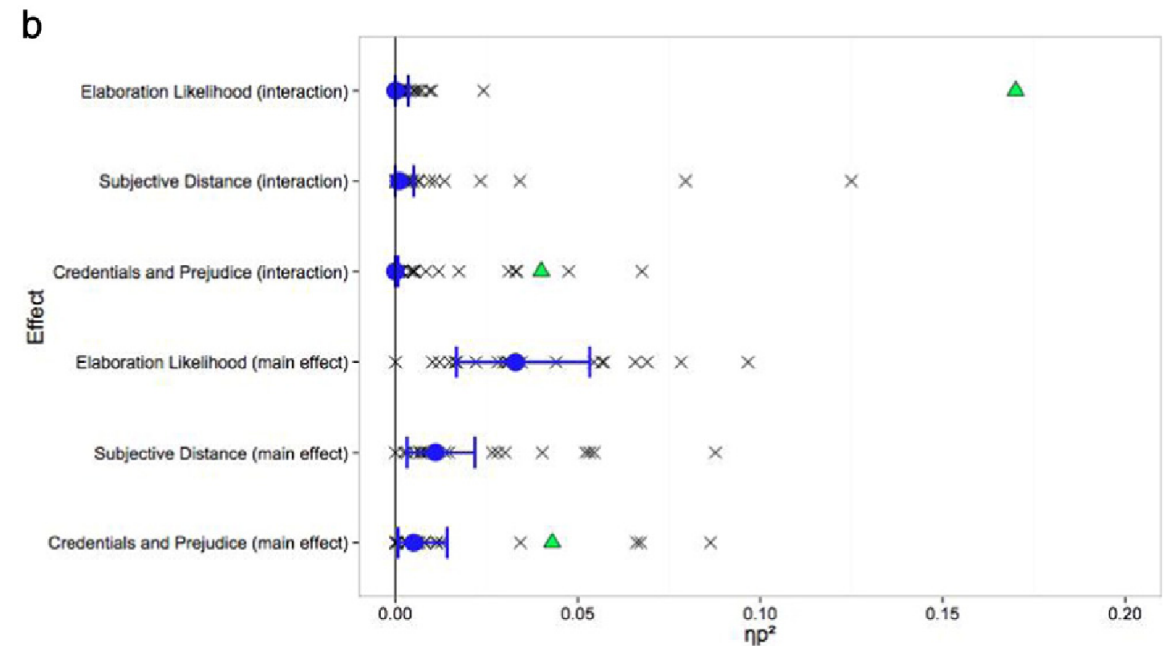
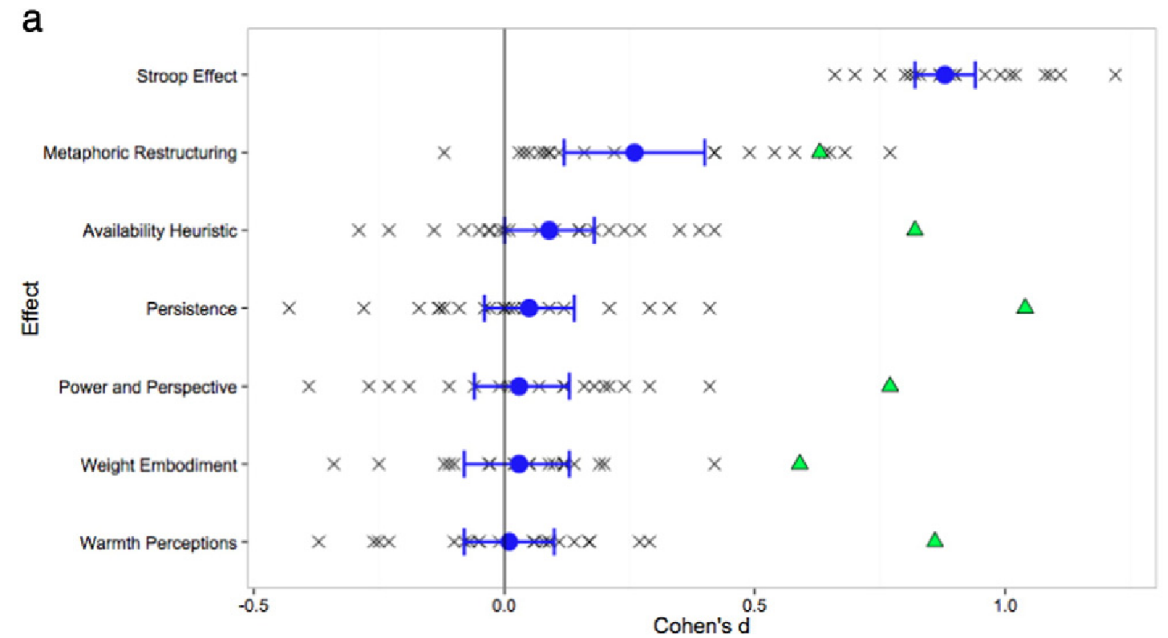
„The spark“ (Open Science Collaboration, 2015)

Many Labs 1 (Klein et al., 2014)

Many Labs 2 (Klein et al., 2018)

Many Labs 3 (Ebersole et al., 2016)

Evaluating participant pool quality across the academic semester via replication



„The spark“ (Open Science Collaboration, 2015)

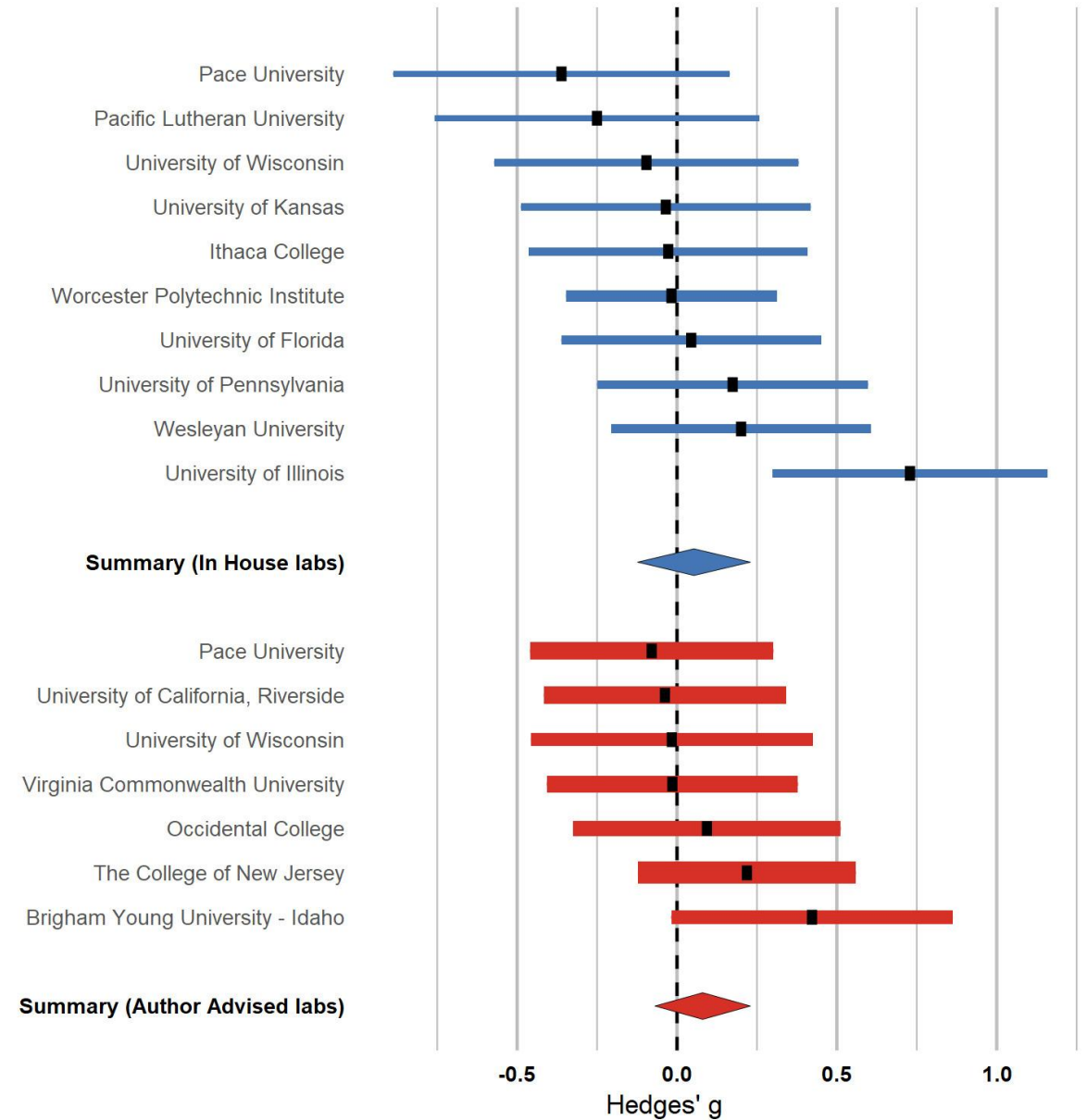
Many Labs 1 (Klein et al., 2014)

Many Labs 2 (Klein et al., 2018)

Many Labs 3 (Ebersole et al., 2016)

Many Labs 4 (Klein et al., 2022)

Failure to Replicate Mortality Salience Effect With and Without Original Author Involvement



Many Labs Project

„The spark“ (Open Science Collaboration, 2015)

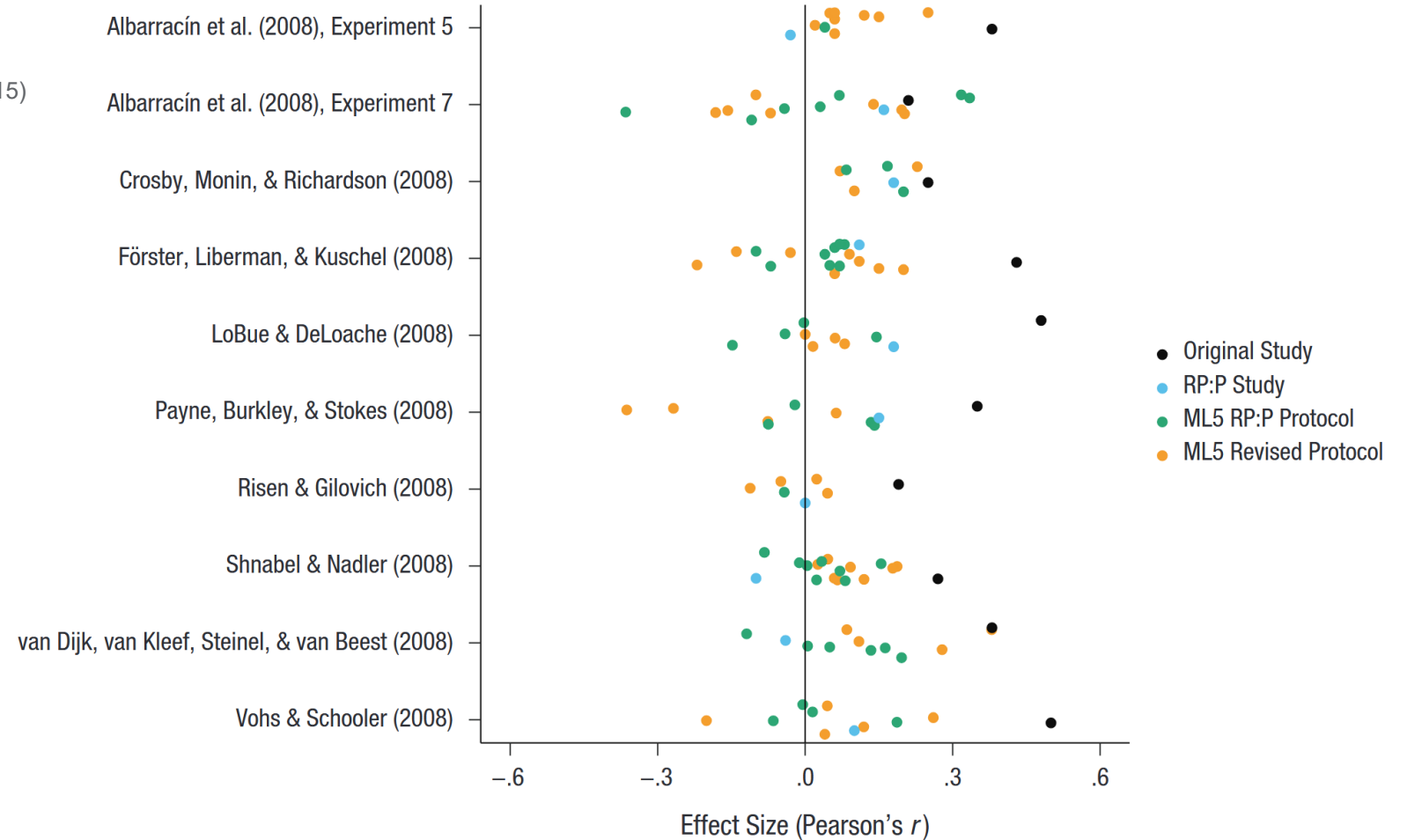
Many Labs 1 (Klein et al., 2014)

Many Labs 2 (Klein et al., 2018)

Many Labs 3 (Ebersole et al., 2016)

Many Labs 4 (Klein et al., 2022)

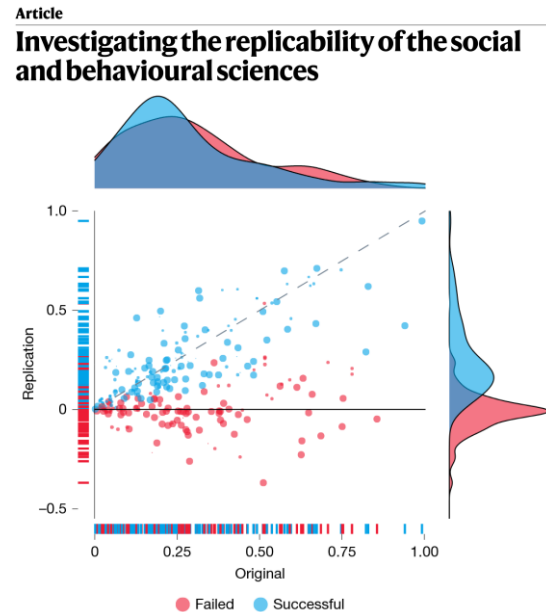
Many Labs 5 (Ebersole et al., 2020)



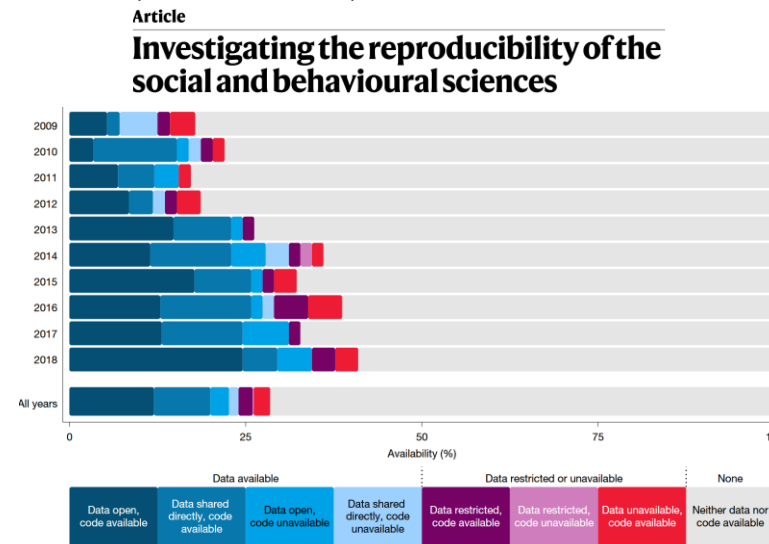
Testing Pre-Data-Collection Peer Review as an
Intervention to Increase Replicability

- The **S**ystematizing **C**onfidence in **O**pen **R**esearch and **E**vidence project developed in the spirit of the Many Labs Projects.
- Recently (2026), three major papers were published in Nature:

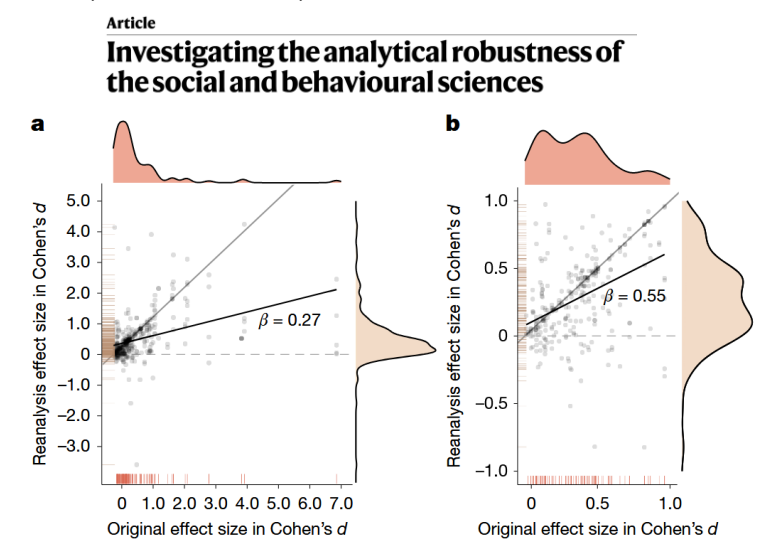
(Tyner et al., 2026)



(Miske et al., 2026)



(Aczel et al., 2026)



Fraud cases and Questionable Research Practices

- In the early 2000s, a series of data fraud cases shocked the scientific community
- Some notable cases from psychology:

Diederik Stapel:

- Prof. in social psychology
- 2011 suspended for fabrication of data used in publications.
- 58 publications were retracted due to data fraud.

Marc Hauser:

- Prof. for evolutionary psychology
- 2011 banned from teaching/resigned from Harvard.
- 2012 found guilty of scientific misconduct due to data fabrication and results manipulation.

Lawrence Sanna:

- Prof. for social psychology
- 2012 resigned from University of Michigan.
- 5 publications were retracted due to „unreliable data“.

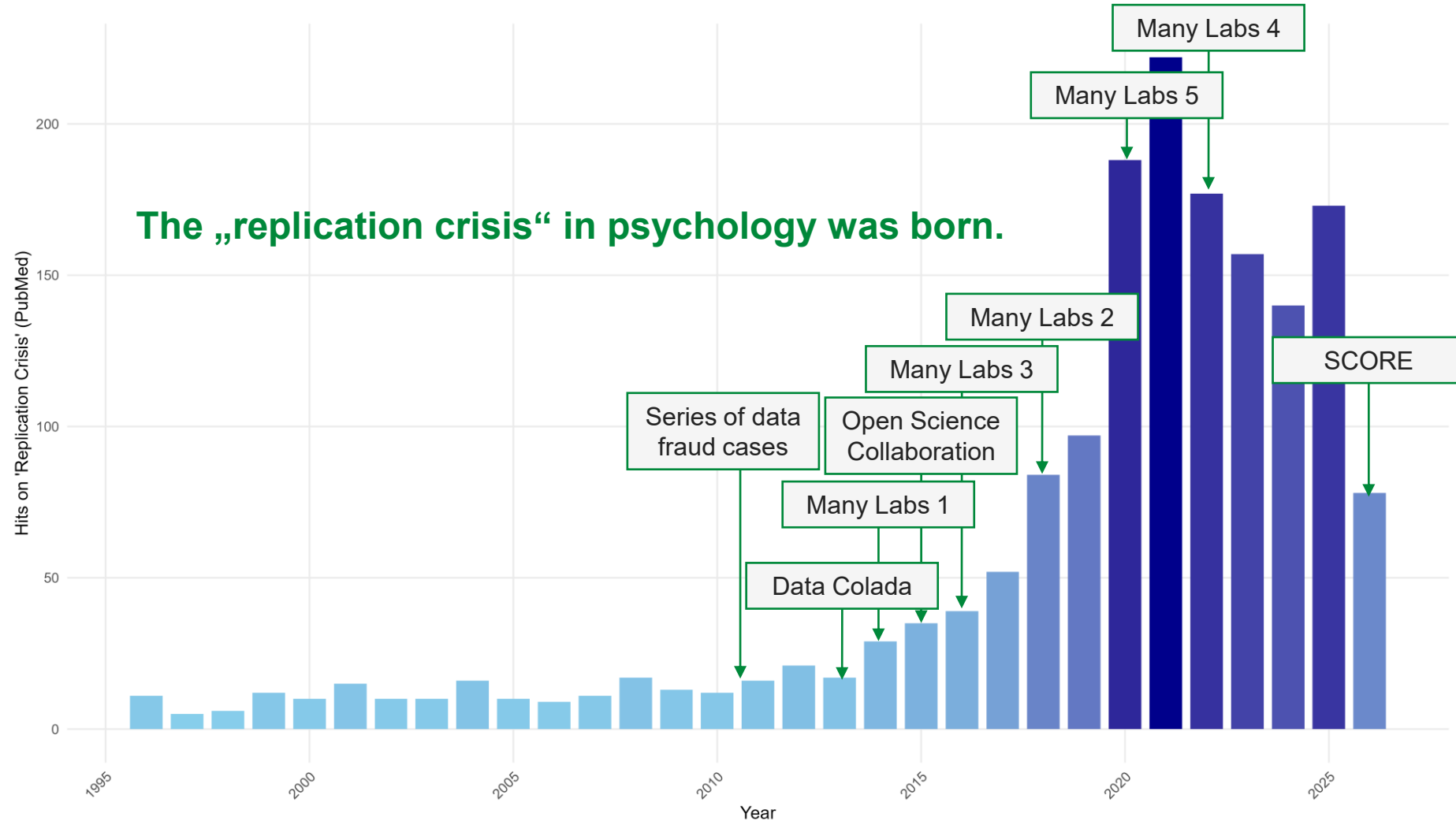
Fraud cases and Questionable Research Practices

- In the same time, the discussion about Questionable Research Practices sparked.
- More than 1000 publications on the topic of QRPs between 2005 and 2015
- The widely known blog for data and data analysis investigations *Data Colada* was founded in 2013.



Thinking about evidence, and vice versa

Crisis narrative



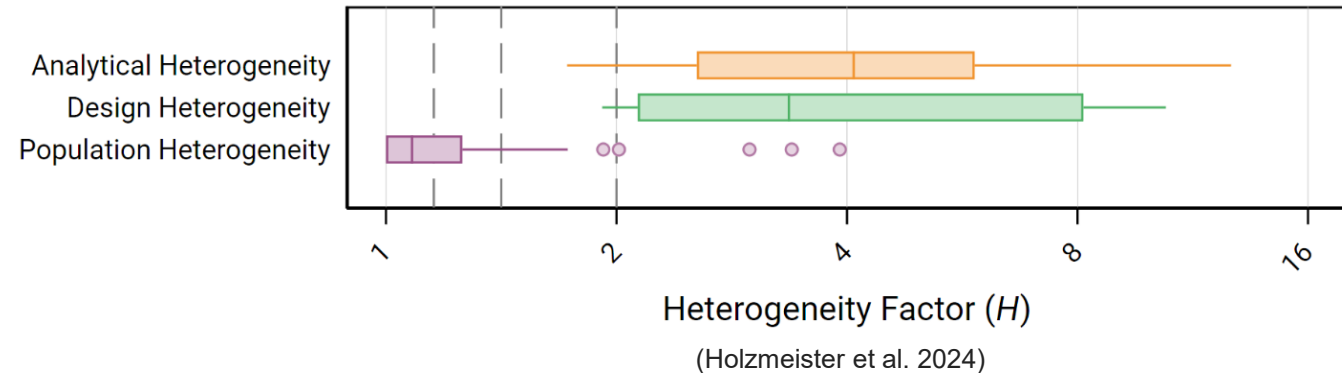
- Numerous grassroots initiatives emerged in response to concerns about replicability.
- The DFG funded the six-year priority programme:
„META-REP: A Meta-scientific Programme to Analyse and Optimise Replicability in the Behavioural, Social and Cognitive Sciences”
- META-REP brings together researchers from psychology, sociology, economics, cognitive science, statistics, machine learning, and philosophy of science
- The programme addresses three central questions:
 1. What is replicability
 2. Why do studies fail to replicate?
 3. How can replicability be improved

1) What exactly is replicability?

- Different terms in different disciplines.

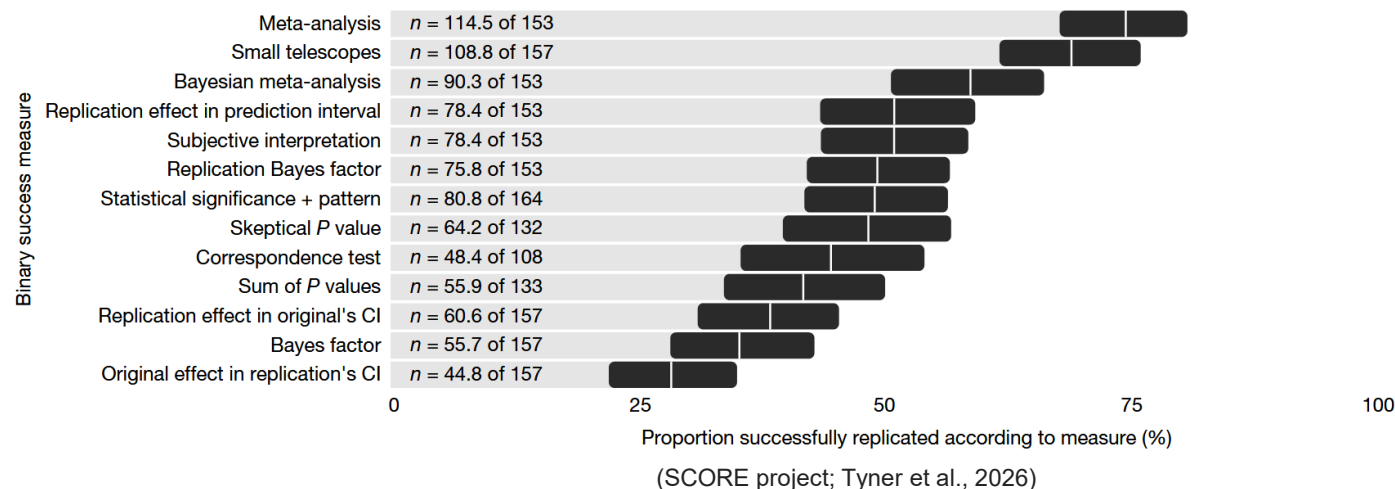
	Same data	Collecting new data
Same materials, methods code etc.	Reproducibility Analysis	Direct Replication
Different materials, methods code etc.	Robustness Analysis	Conceptual Replication

- How much deviation between a primary/original study and a replication study is allowed?
- Which effects are introduced by variations?
- What about generalizability of scientific findings?



1) What exactly is replicability?

- What exactly defines a succesful replication?
 - Observing statistical significance and direction of the original effect? (Open Science Collaboration, 2015)
 - Observing confidence interval of the original effect size?(Patil et al., 2016, Verhagen & Wagenmakers, 2014)
 - Bayesian inference?
 - A mixture of qualitative and quantitative measures? (LeBel et al., 2018)

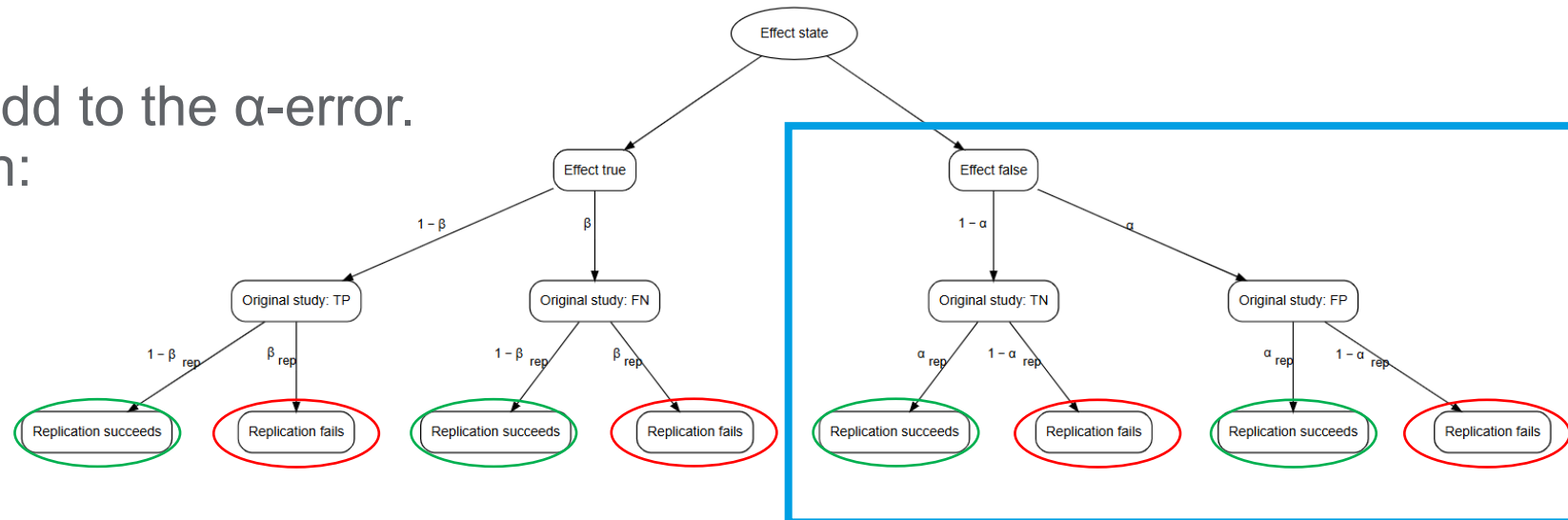


2) Why do studies fail to replicate?

- Two reasons are discussed as the cause for failed replications:
 1. Inflated false-positive rates.
 2. Context dependencies of effects.

2) Why do studies fail to replicate? Inflated false-positive rates.

- Anything that unconditionally increases the chance of “finding a significant effect” also increases the risk of falsely accepting non-effects as significant.
 - Larger α -error (Ioannidis, 2005; Simmons et al. 2012)
- In Null Hypothesis Significance Testing (NHST), we always accept a base α -error, the critical value (often .05).
- But also other causes can add to the α -error.
You can distinguish between:
 - Individual causes.
 - Systemic causes.

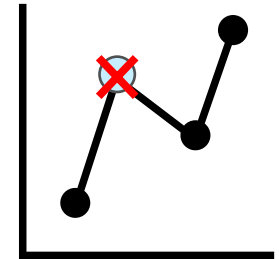


2) Why do studies fail to replicate? Inflated false-positive rates.

- **Individual level causes** leading to inflated false-positive rates
 - Questionable research practices(QRPs). (John et al., 2012; Bakker et al., 2012)
 - Data cherry-picking

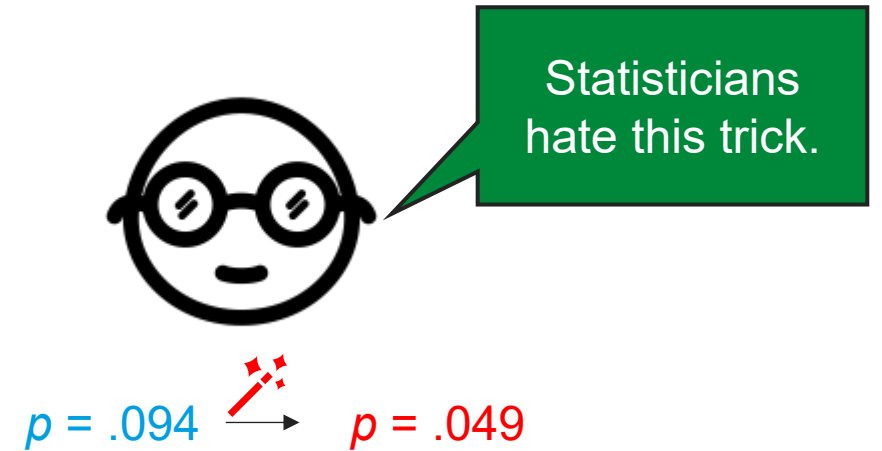


I hate this
participant's
data



2) Why do studies fail to replicate? Inflated false-positive rates.

- **Individual level causes** leading to inflated false-positive rates
 - Questionable research practices(QRPs). (John et al., 2012; Bakker et al., 2012)
 - Data cherry-picking
 - *P*-hacking



2) Why do studies fail to replicate? Inflated false-positive rates.

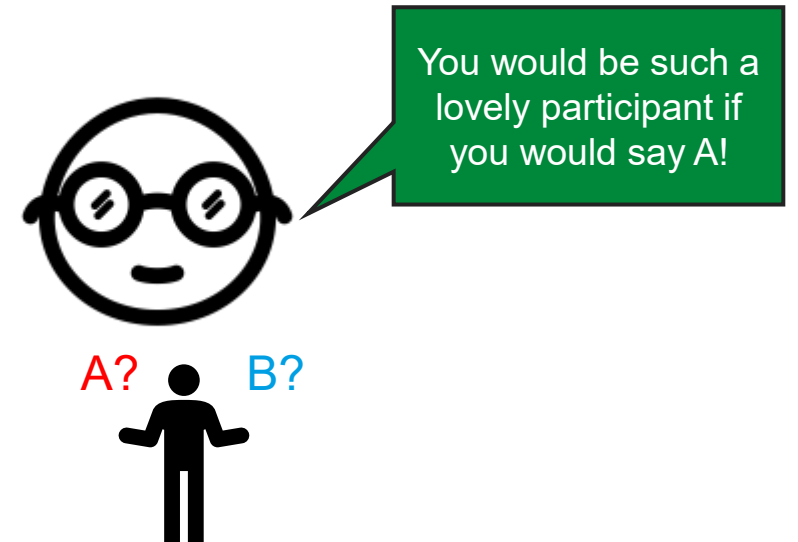
- **Individual level causes** leading to inflated false-positive rates
 - Questionable research practices(QRPs). (John et al., 2012; Bakker et al., 2012)
 - Data cherry-picking
 - *P*-hacking
 - HARKing



Wait, what are
we testing
again?

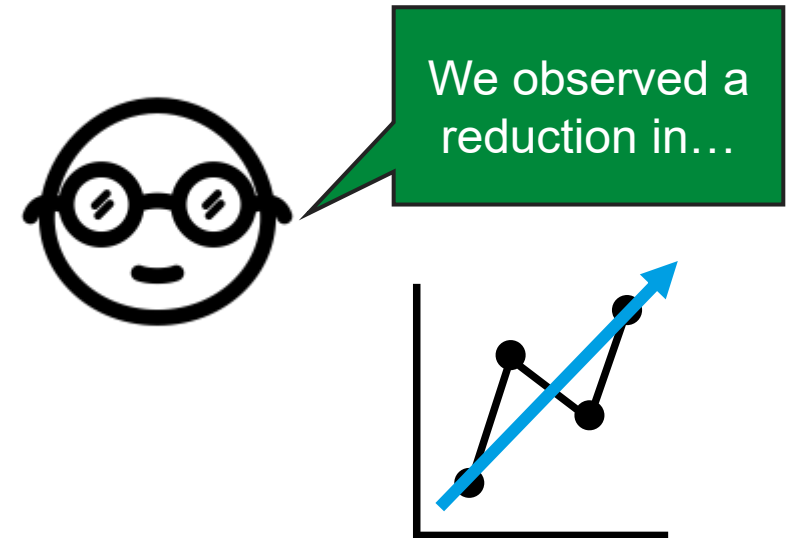
2) Why do studies fail to replicate? Inflated false-positive rates.

- **Individual level causes** leading to inflated false-positive rates
 - Questionable research practices(QRPs). (John et al., 2012; Bakker et al., 2012)
 - Data cherry-picking
 - *P*-hacking
 - HARKing
 - Inducing hypothesis-aligned biases in the participants



2) Why do studies fail to replicate? Inflated false-positive rates.

- **Individual level causes** leading to inflated false-positive rates
 - Questionable research practices(QRPs). (John et al., 2012; Bakker et al., 2012)
 - Data cherry-picking
 - *P*-hacking
 - HARKing
 - Inducing hypothesis-aligned biases in the participants
 - Drawing misleading/incorrect conclusions



2) Why do studies fail to replicate? Inflated false-positive rates.

- **Individual level causes** leading to inflated false-positive rates
 - Questionable research practices(QRPs). (John et al., 2012; Bakker et al., 2012)
 - Data cherry-picking
 - *P*-hacking
 - HARKing
 - Inducing hypothesis-aligned biases in the participants
 - Drawing misleading/incorrect conclusions
 - Selection of inappropriate methods. (Smaldino & McElreath, 2016)



My supervisor, my
friends, my
reviewers, we are all
doing it!

2) Why do studies fail to replicate? Inflated false-positive rates.

- **Individual level causes** leading to inflated false-positive rates
 - Questionable research practices(QRPs). (John et al., 2012; Bakker et al., 2012)
 - Data cherry-picking
 - *P*-hacking
 - HARKing
 - Inducing hypothesis-aligned biases in the participants
 - Drawing misleading/incorrect conclusions
 - Selection of inappropriate methods. (Smaldino & McElreath, 2016)
 - Motivated reasoning. (Ioannidis et al., 2014)



I really want/need
significant results
in this study!

2) Why do studies fail to replicate? Inflated false-positive rates.

- Systemic level causes leading to inflated false-positive rates
 - Scientists are incentivized to produce impactful publications.
 - PhDs want to finalize their dissertation project.
 - Postdocs want a tenured position.
 - Senior researchers want money to pay their employees and research.
 -



I need to publish
or I will perish.

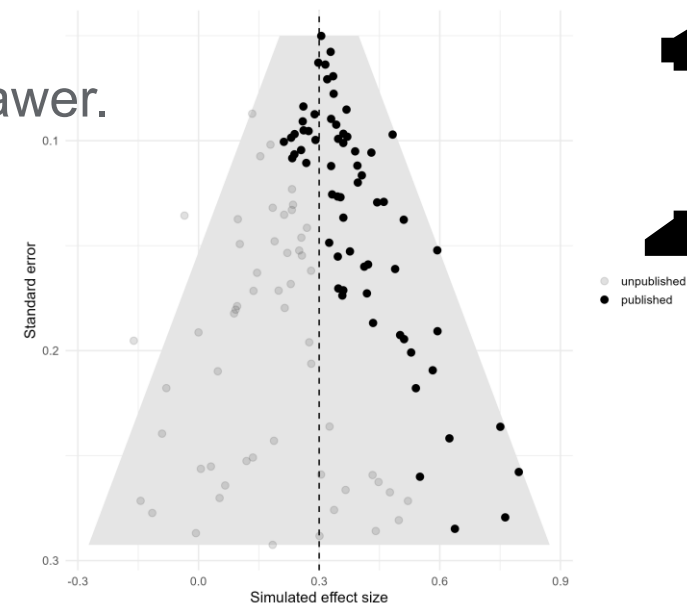
2) Why do studies fail to replicate? Inflated false-positive rates.

- Systemic level causes leading to inflated false-positive rates
- Scientists are incentivized to produce impactful publications.
 - PhDs want to finalize their dissertation project.
 - Postdocs want a tenured position.
 - Senior researchers want money to pay their employees and research.
- Journals (often) want novelty → new results
 - Null- and inconclusive results remain in the file drawer.
 - Accumulation of published false-positive results.



I need to publish
or I will perish.

We will publish all
the new,
significant effects.



„What is Good for Scientists?“

VS

„What is Good for Science?“

(Nosek et al., 2012)

2) Why do studies fail to replicate? A (simple) theoretical model

Let's run 1000 studies with

- critical $\alpha = .05$
 - testpower of $\beta = .8$
- 500 of the studies yield significant results.
→ 500 are insignificant.

600 effects are a priori existing
400 effects are a priori non-existing

True positives: 480	$\beta = .80$ (test power): 120
	True negatives: 380
$\alpha = .05$ (NHST): 20	

2) Why do studies fail to replicate? A (simple) theoretical model

Now, we run for each study an identical replication ($\alpha = .05$; $\beta = .8$)

True positives: 480	$\beta = .80$ (test power): 120
	True negatives: 380
$\alpha = .05$ (NHST): 20	



TP not replicated: 96	FN replicated: 24
TP replicated: 384	FN not replicated: 96
	TN replicated: 361
FP not replicated: 19	TN not replicated: 19

Successful replications: 770
Failed replications: 230

Replication rate: 77%

2) Why do studies fail to replicate? A (simple) theoretical model

Unfortunately, that scenario was quite optimistic.

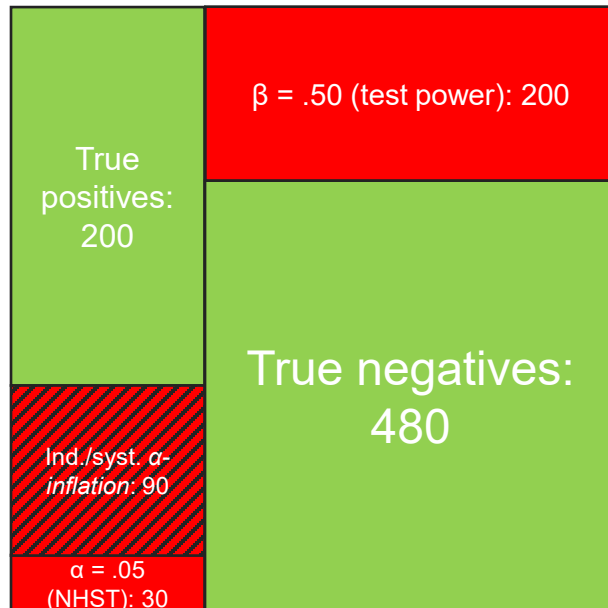
Maybe more realistic:

Critical $\alpha = .05$ + α inflation due to individual/systemic factors $\alpha = .15$ (total $\alpha = .2$)

testpower of $\beta = .6$, but 80% of the not significant results, are in the file drawer.

400 effects are a priori existing

600 effects are a priori non-existing



2) Why do studies fail to replicate? A (simple) theoretical model

Unfortunately, that scenario was quite optimistic.

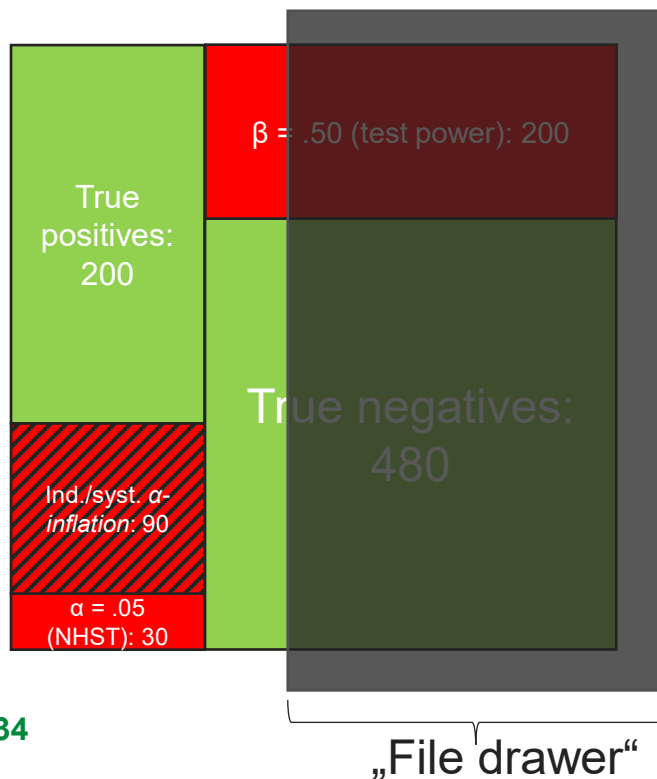
Maybe more realistic:

Critical $\alpha = .05$ + α inflation due to individual/systemic factors $\alpha = .15$ (total $\alpha = .2$)

testpower of $\beta = .6$, but 80% of the not significant results, are in the file drawer.

400 effects are a priori existing

600 effects are a priori non-existing



2) Why do studies fail to replicate? A (simple) theoretical model

Unfortunately, that scenario was quite optimistic.

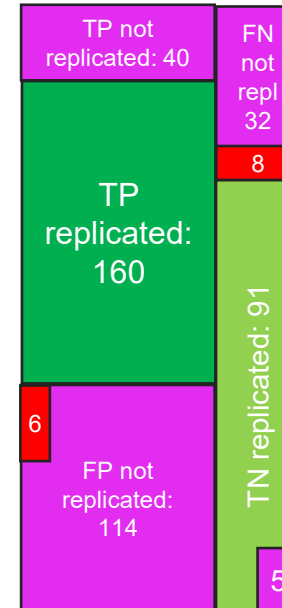
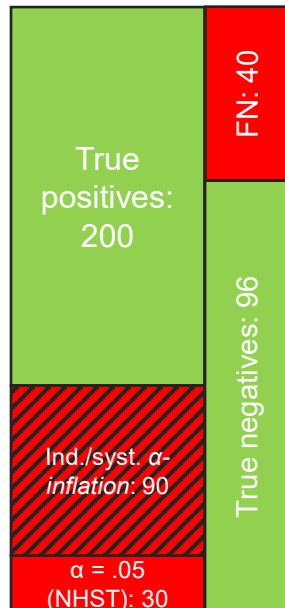
Maybe more realistic:

Critical $\alpha = .05$ + α inflation due to individual/systemic factors $\alpha = .15$ (total $\alpha = .2$)

testpower of $\beta = .6$, but 80% of the not significant results, are in the file drawer.

400 effects are a priori existing

600 effects are a priori non-existing



Successful replications: 265

Failed replications: 191

Replication rate: 58%

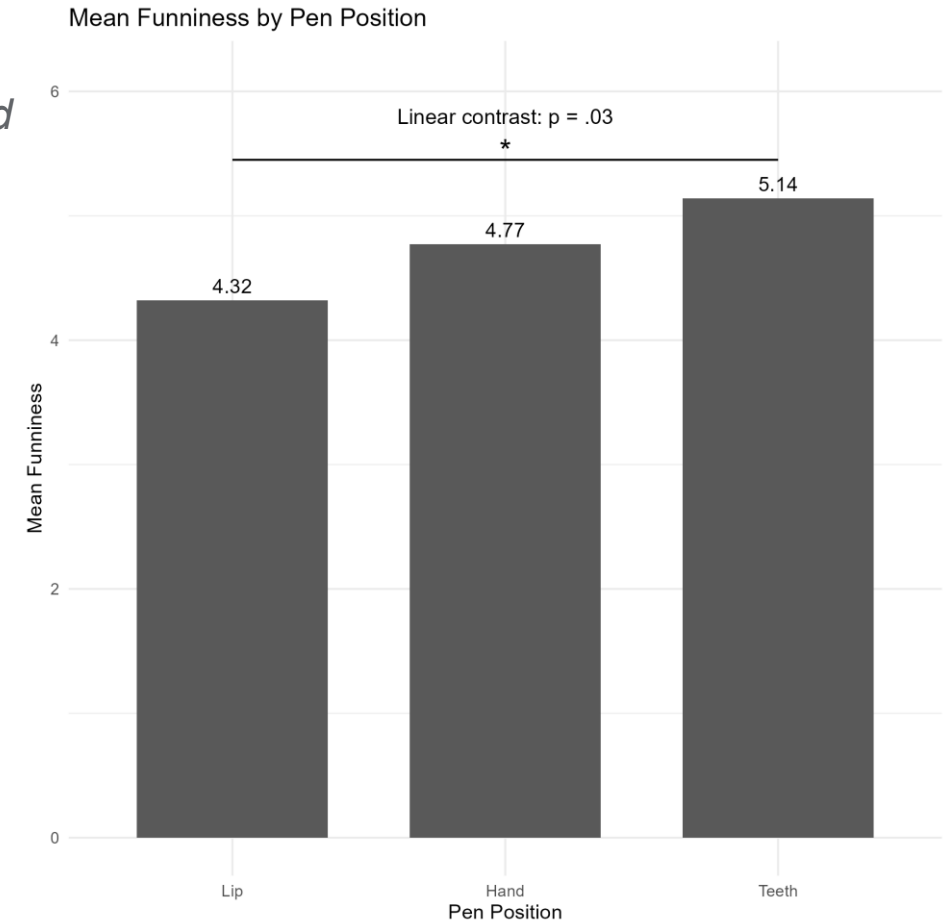
2) Why do studies fail to replicate? Context dependencies of effects.

- So far, we have only talked about error-like factors that can lead to a study not replicating.
- But what if a failed replication itself can highlight gaps or boundaries in a theory?

2) Why do studies fail to replicate? Context dependencies of effects.

The case of the „facial feedback“ effect.

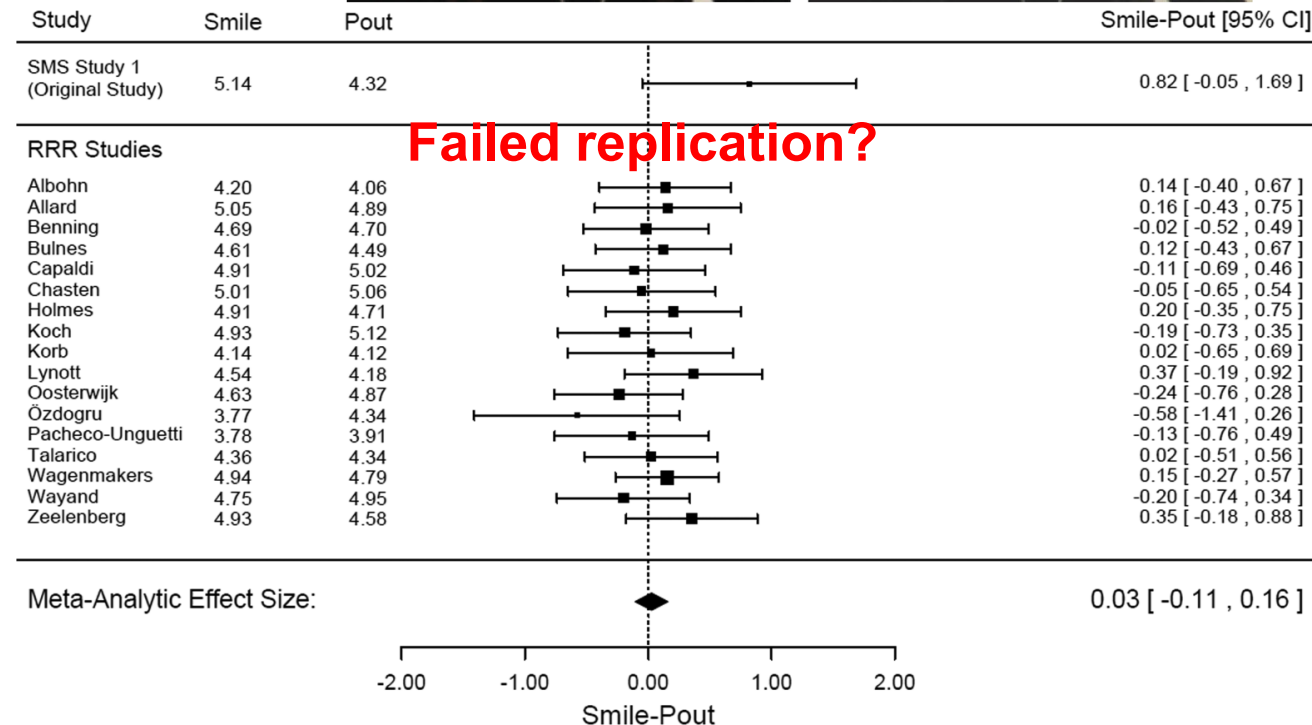
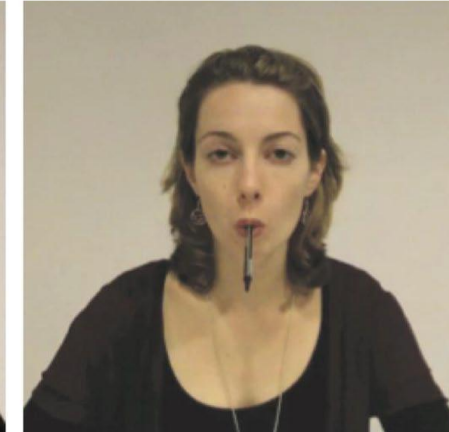
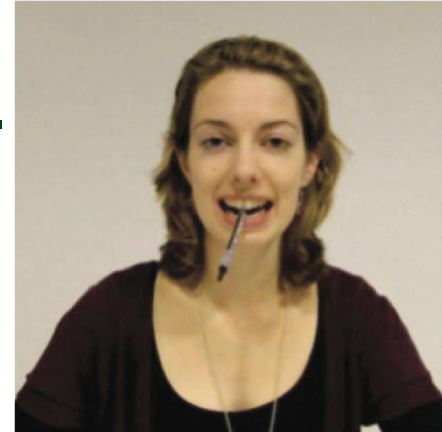
- Original effect: Strack et al., 1988
- *It was hypothesized that more or less positive affective states, as reflected in the experience of humor, would be influenced by muscle contractions associated with smiling. Specifically, potentially humorous stimuli should be rated least funny when the muscles associated with smiling are inhibited (lips condition), but should be rated most funny when this muscular activity is facilitated (teeth condition).*



2) Why do studies fail to replicate? Context dependencies of effects.

The case of the „facial feedback“ effect.

Direct replication in 17 independent labs:
Wagenmakers et al., 2016



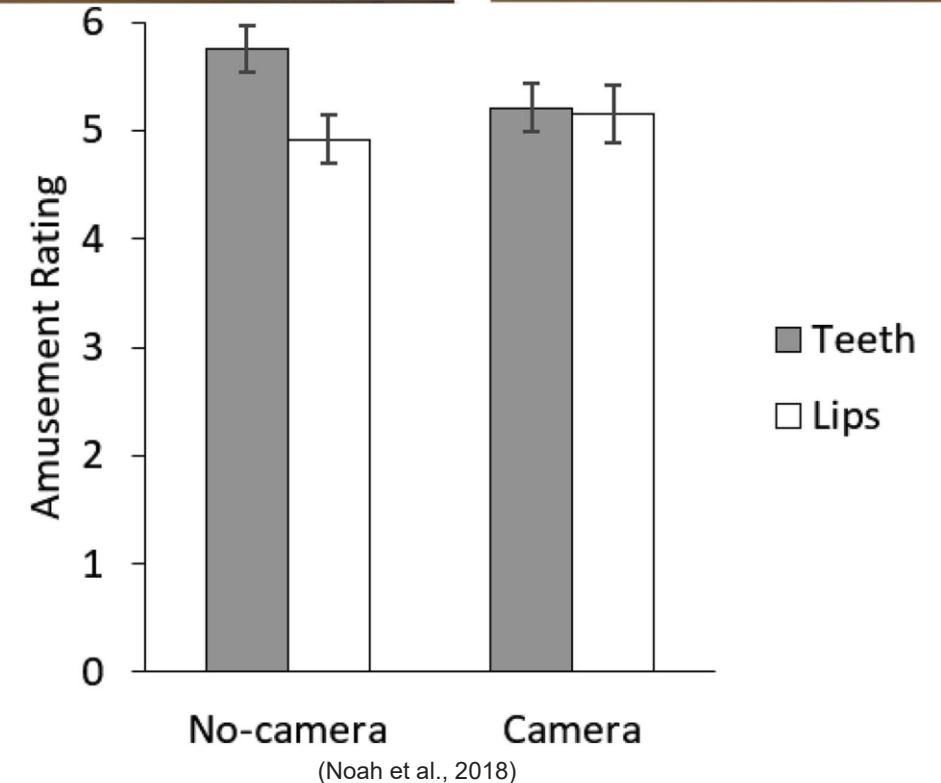
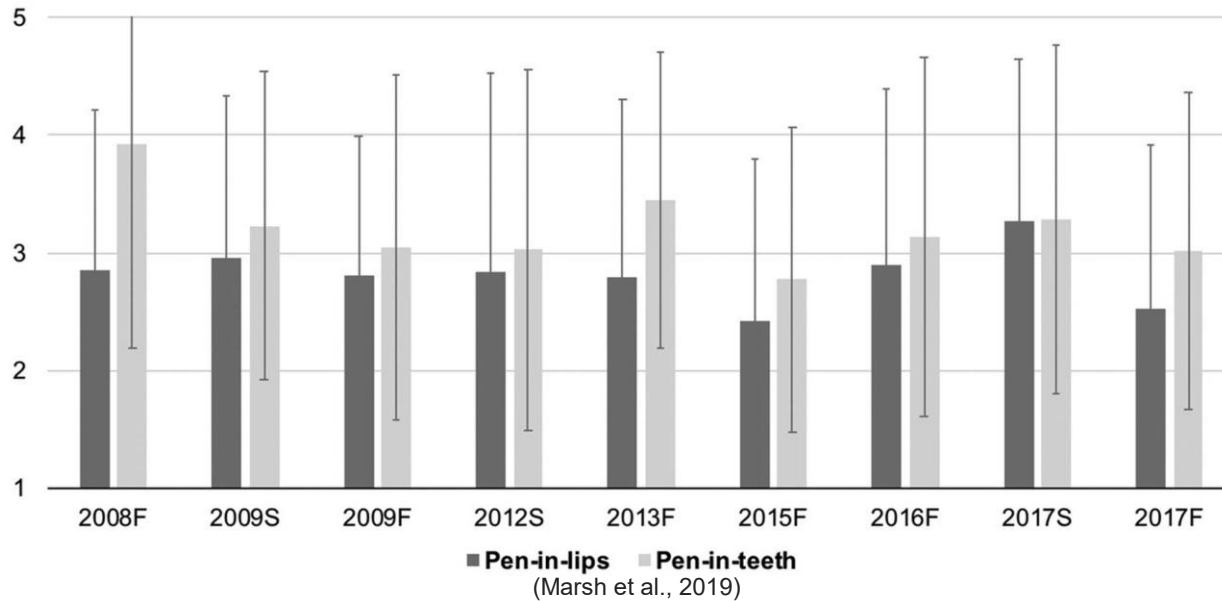
2) Why do studies fail to replicate? Context dependencies of effects.

The case of the „facial feedback“ effect.

More replications ruling out these concerns:

- Noah et al. (2018)
- Marsh et al. (2019)

→ Small-sized effect

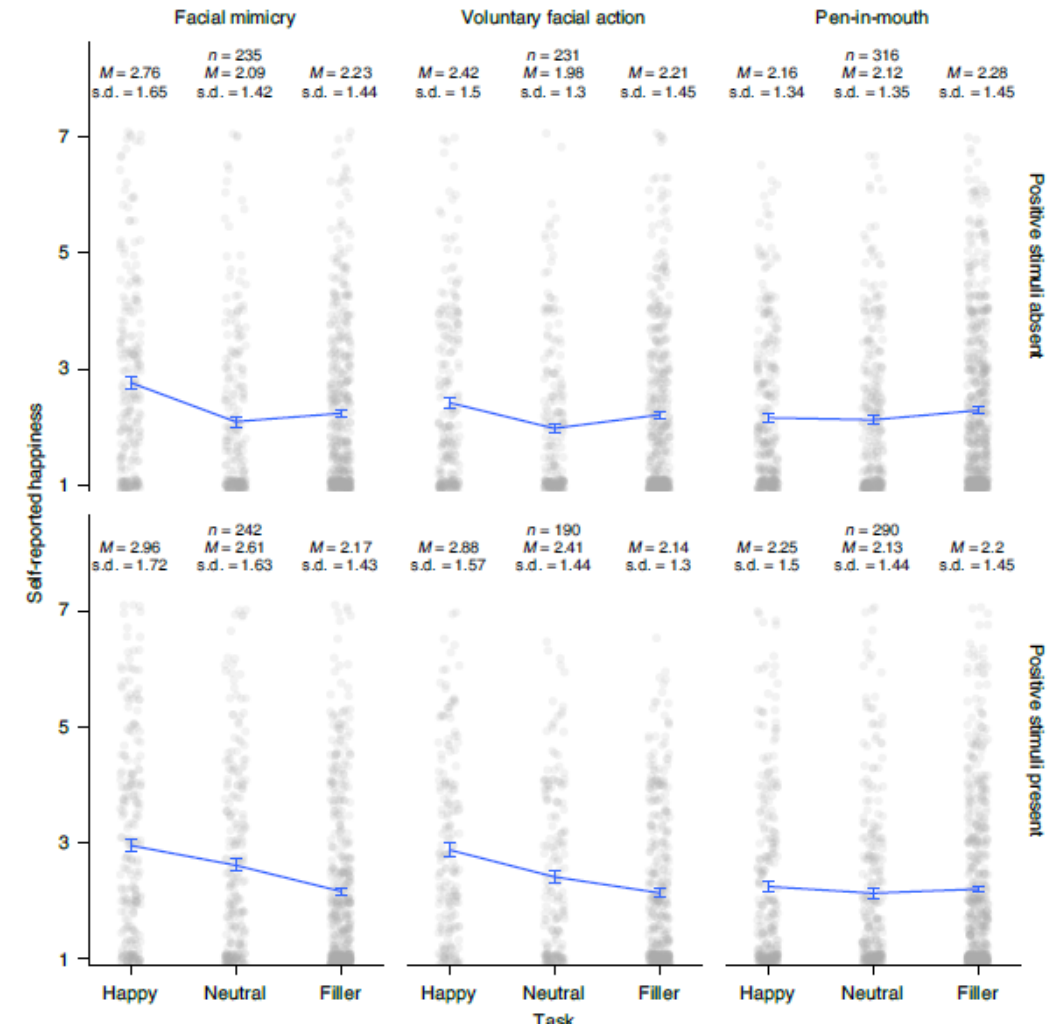


2) Why do studies fail to replicate? Context dependencies of effects.

The case of the „facial feedback“ effect.

There is also a recent multi-lab replication project on the topic of facial feedback:

Coles et al. (2022)



2) Why do studies fail to replicate? Context dependencies of effects.

- The argument of context-effects being the reason for low replicability rates, is on the one hand obvious but also complicated.
- The problem with the argument is that you might argue that each failed replication results from context-differences between the primary and the replication study
 - This could be used as a strategic argument to shield studies against criticism.
 - At the same time it would make the studies unfalsifiable and thus unscientific.
- So what counts as a context?

Does the context moderate the effect in a way that is theoretically meaningful?

or

Does the context only moderate the effect on an operational or measurement level?

2) Why do studies fail to replicate? Context dependencies of effects.

Context characteristic	Theoretically meaningful context moderator	Operational / measurement-level moderation
1. Sample characteristics	<i>A social conformity effect is stronger among adolescents than adults because peer approval is more central to adolescents' social motivation.</i> → Age is a theoretically meaningful.	<i>The effect appears weaker in an older sample because older participants respond slower in an unfamiliar computer setting.</i> → Age just affects the measurement.
2. Psychometric characteristics of the independent variable	<i>Arousal can be induced through physical activity or by experiencing emotional stimuli.</i> → On first sight, similar manipulations can activate different processes.	<i>One study uses all stimuli of a sample, the other task only a subset. Both sets have identical properties.</i> → The stimuli might not be validated properly.
3. Psychometric characteristics of the dependent variable	<i>Arousal can be measured through skin conductance or pupillometry but on a different time scale.</i> → The time scale might highlight differences in the underlying process.	<i>One study uses the full questionnaire, the other study the short version.</i> → The measurement instrument might not be validated properly.
4. Setting characteristics	<i>The use of camera makes participants more self aware thus affecting their mimic.</i> → Being filmed can be theoretically meaningful.	<i>Only in a lab studies, participants comply to the experimenter demands. The effect does not appear in an online setting.</i> → The lab/online condition is not theoretically relevant.

2) Why do studies fail to replicate? Context dependencies of effects.

- The scientific discussion on what qualifies as a theoretically meaningful context and how large are these context effects is ongoing.
- Posthoc analysis of the studies of the first Many Labs project
 - Experts judgments on the context-sensitivity of the effects seem to predict their replicability. (van Bavel et al. 2016; critical comment in Yoel, 2016)
- However, in the second Many Labs project, found no substantial effect-moderation by analyzed contexts. (Klein et al., 2018)

3) What can be done to achieve acceptable replication rates?

What is an acceptable replication rate?

- To achieve 100% replication rate, scientific studies would only confirm what is already known (extremely high prior odds).

(Nosek et al., 2022)

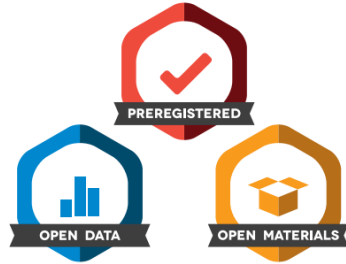
Discipline	Replication attempt statistical significance and pattern		Median Pearson's r effect size (s.d.)	
	Counts	Percentage	Original estimate	Replication estimate
Business	17.0 of 36	47.2	0.24 (0.12)	0.10 (0.13)
Economics	10.2 of 24	42.5	0.28 (0.24)	0.13 (0.20)
Education	8.2 of 13	63.1	0.15 (0.28)	0.11 (0.10)
Political science	7.8 of 15	52.0	0.16 (0.22)	0.05 (0.16)
Psychology	28.4 of 58	49.0	0.29 (0.22)	0.11 (0.20)
Sociology	9.2 of 18	51.1	0.10 (0.16)	0.03 (0.17)

(Tyner et al., 2026)

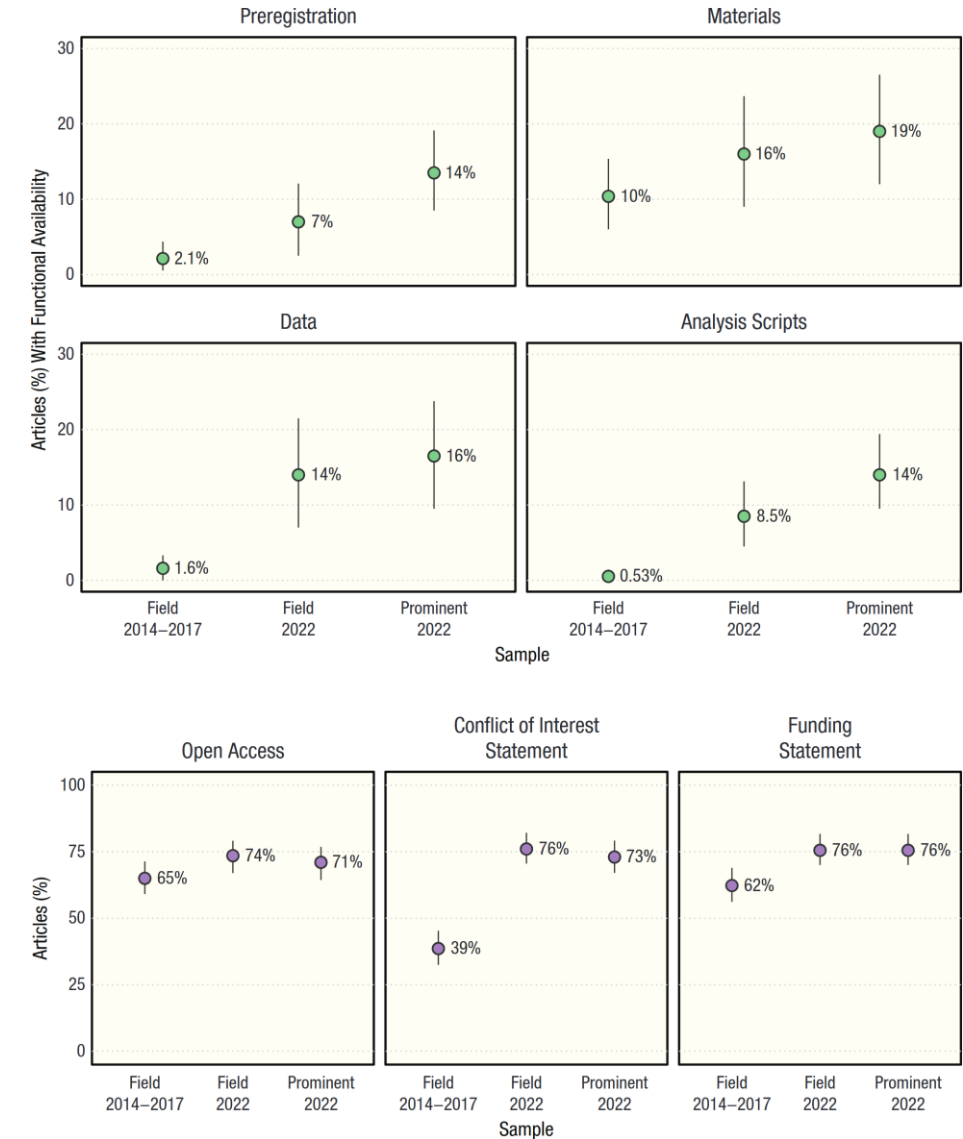
- But what is an acceptable replication rate?
 - The question is still up for debate. (Nosek et al., 2022; Tyner et al., 2026; Rubin 2026)
 - Replication rates differ between fields.
 - Simulation studies indicate that as long public replications are performed, even under low-replicability conditions knowledge gain is possible. (Lewandowsky & Oberauer, 2020)

3) What can be done to achieve acceptable replication rates?

Open Science



- Is it enough if, we all adopt *Open Science Practices*?
- Unfortunately, it is more complicated.
 - Primarily, *Open Science Practices* enable scientific work to be replicated.
 - However, it remains unclear if *Open Science* has better replication rates.
 - Also, it does not change the lack of incentives to replicate others' work.
- But most importantly, we are far away from field-wide application *Open Science Practices*.



(Hardwicke et al., 2024)

3) What can be done to achieve acceptable replication rates?

Improving theories

- Better links between theories and derived research questions can improve
 - the prior probability of a research question to be true.
 - the selection of appropriate research instruments.
- Distinguishing between *discovery-oriented research* and *theory-testing research*
 - *Discovery-oriented research* (less-developed theories, low prior probabilities) should use large sample sizes and/or replications to establish new findings.
 - *Theory-testing research* (established theories, higher prior probabilities) should aim for precise predictions with the ideal aim of formalization.

(Nosek et al. 2022)

(Oberauer & Lewandowsky, 2019)

3) What can be done to achieve acceptable replication rates?

Increasing „effort“

- Improving methodology, such as increasing observations, stronger measures and manipulations etc. (Vazire, 2018)
- Applying different inference criteria. (Lakens et al., 2018)
- Seriously consider alternatives (e.g., robustness checks, cross-validations, and internal replications). (Nosek et al. 2022)
- Moving towards large-scale research projects with multiple populations, method selections, and analysis approaches. (Holzmeister et al., 2024)
- But
 - Asking for more effortful procedures for those who are willing does not remove the social dilemma „good science vs succesful scientist“. (Abele-Brehm et al., 2019)
 - Increasing focus on reducing false-positives might produce more false-negative findings. (Fiedler et al., 2012)

Brief excursion

DFG Priority Programme (Schwerpunktprogramm, SPP)

- The Priority Programmes (funded by the *Deutsche Forschungsgemeinschaft*) aim to give clear impetus to the advancement of science through coordinated multi-location funding for important new topics.
- How does it work?
 - a. A group of established researchers develops a scientific concept for the targeted research field and submit it to the DFG
 - b. If approved, the DFG establishes the program (2x 3years)
 - c. A coordination team organizes the collaboration within the programme
 - d. Researchers from German institutions can apply with individual research projects.
 - e. These projects are evaluated through the regular competitive DFG process→ The application for an SPP does not start with a predetermined group of reserachers.

Programme Comittee



Mario Gollwitzer
LMU



Felix Schönbrodt
LMU



Susann Fiedler
WU Wien



Tina Lonsdorf
Uni Bielefeld



Katrin Auspurg
LMU

Programme Coordination



Moritz Schiltenswolf
LMU



Maximilian Frank
LMU

META-REP

First funding phase (2022-2024)

Project 1: *A Large-scale Collaborative Assessment of the Replicability and Robustness of EEG Research*

PIs: Prof. Dr. Niko Busch and Dr. Yuri Pavlov

ECRs: Elena Cesnaite

[Video-Introduction](#)

Project 2: *Conceptual Replications - Guidelines for Implementation and Factors Influencing Replicability across Different Fields in Psychology*

PIs: Prof. Dr. Marie-Ann Sengewald, Prof. Dr. Anne Gast, Prof. Dr. Steffi Pohl, and Dr. Mathias Twardawski

ECRs: Johanna Höhs and Dennis Kondzic

[Video-Introduction](#)

META-REP

First funding phase (2022-2024)

Project 3: *Cognitive Modelling: Blessing or Curse for Replication in Psychology?*

PIs: : Prof. Dr. Michael Zehetleitner & Dr. Manuel Rausch

ECRs: Cem Tabakci

[Video-Introduction](#)

Project 4: *Enhancing the Robustness of Observational Social Science Research by Computational Multi-Model Analyses*

PIs: Prof. Dr. Katrin Auspurg & Dr. Andreas Schneck

ECRs: Daniel Krähmer & Laura Schächtele

[Video-Introduction](#)

META-REP

First funding phase (2022-2024)

Project 5: *Explaining Heterogeneity in Direct Replications: Just Methodological Artefacts?*

PIs: Dr. Frank Renkewitz & Prof. Dr. Moritz Heene

ECRs: Lukas Beinhauer & Jens Funderich & Maximilian Frank

[Video-Introduction](#)

Project 6: *How an Academic System Can Achieve a Trustworthy Knowledge Base: Analyzing Reform Proposals in an Agent-based Modeling Approach*

PIs: Prof. Dr. Felix Schönbrodt & Prof. Dr. Richard McElreath & Dr. Filip Melinscak

ECRs: Florian Kohrt

[Video-Introduction](#)

META-REP

First funding phase (2022-2024)

Project 7: *METEOR – MastEring ThE OppRessive number of forking paths unfolded by noisy and complex neural data*

PIs: Prof. Dr. Andrea Hildebrandt & Prof. Dr. Stefan Debener & Prof. Dr. Christiane Thiel & Dr. Carsten Gießing

ECRs: Nadine Jacobsen & Daniel Kristanto

Project 8: *Robustness and Replicability in Economics (R2E)*

PIs: Prof. Dr. Anna Dreber Almenberg & Prof. Dr. Nathan Fiala & Prof. Dr. Magnus Johannesson & Prof. Dr. Jörg Peters & Dr. Mandy Malan;

ECRs: : Florian Neubauer & Julian Rose

[Video-Introduction](#)

[Video-Introduction](#)

META-REP

First funding phase (2022-2024)

Project 9: *Replicating Failed Replications: A Validity-based Framework for Predicting Replication Success*

PIs: : Dr. Marc Jekel

ECRs: Patrick Smela

Project 10: *Standardization of Behavior Research Methods as a Determinant of Replicability*

PIs: Prof. Dr. Malte Elson & Dr. Ruben Arslan

ECRs: Ian Hussey & Taym Alsalti

[Video-Introduction](#)

[Video-Introduction](#)

META-REP

First funding phase (2022-2024)

Project 11: *The Reproducibility and Robustness of Secondary Analyses in Educational Research: The Role of Publication Bias and Researcher Degree of Freedom*

PIs: Dr. Malte Jansen & Dr. Aleksander Kocaj

ECRs: Aishvarya Aravindan Rajagopal

Project 12: *The Role of Measurement in the Replicability of Empirical Findings*

PIs: Prof. Dr. Eunike Wetzel

ECRs: : Caroline Böhm & Steven Bißantz

[Video-Introduction](#)

[Video-Introduction](#)

META-REP

First funding phase (2022-2024)

Project 13: *The Role of Theory in Understanding and Resolving the Reliability Crisis*

PIs: Dr. Nate Breznau

ECRs: Hung Nguyen

Project 14: *Towards a Falsifiable Framework for Theories in Psychological Science*
PIs: Dr. Xenia Schmalz

ECRs: Tatiana Logvinenko & Yi Leung

[Video-Introduction](#)

[Video-Introduction](#)

META-REP

First funding phase (2022-2024)

Project 15: *What Defines and Affects Replicability
in Computational Communication Science?*

PIs: : Dr. Johannes Breuer & Prof. Dr. Mario Haim

ECRs: Philipp Knöpfle

[Video-Introduction](#)

META-REP

Second funding phase (2025-2027)

Project 16: *Applying Principles from the Validity-Based Framework of Replication in Metastudies to Test Causal Relations*

PIs: Dr. Marc Jeke

ECRs: : Patrick Smela

Project 17: *Facing the replication crisis in machine learning modeling*

PIs: Prof. Dr. Ulrich Schroeders & Dr. Kristin Jankowsky

ECRs: Alan Hernes

[Video-Introduction](#)

[Video-Introduction](#)

META-REP

Second funding phase (2025-2027)

Project 18: *A principled Bayesian workflow to study robust individual differences in cognitive science*

PIs: Prof. Julia M. Haaf

ECRs: Anne Giacobello

[Video-Introduction](#)

Project 19: *Fostering Proactive Replicability in Computational Communication Science via Frontloading Effort and Automating Protocols*

PIs: Dr. Johannes Breuer & Prof. Dr. Mario Haim

ECRs: Philipp Knöpfle & Xinyue Zhao

[Video-Introduction](#)

META-REP

Second funding phase (2025-2027)

Project 20: *Addressing the generalizability crisis with large language models*

PIs: : Dr. Dirk Wulff

ECRs: Valentin Kriegmair & Taisiia Tikhomirova

Project 21: *METEOR 2.0 – MastEring community knowledge-driven robusTnEss analysis in cOgnitive neuroscience*

PIs: : Prof. Dr. Stefan Debener, Dr. Carsten Gießing, Prof. Dr. Andrea Hildebrandt & Prof. Dr. Christiane M. Thiel

ECRs: Sana Hassan Imam & Julius Welzel

META-REP

Second funding phase (2025-2027)

Project 22: *Models, Measures, Moderators - Exploring and Explaining Heterogeneity in Psychological Replications*

PIs: Prof. Dr. Moritz Heene & Frank Renkewitz

ECRs: : Maximilian Frank & Teresa Rupperecht

[Video-Introduction](#)

Project 23: *Enhancing Reproducibility and Robustness of Observational Social Science Research*

PIs: : Prof. Dr. Katrin Auspurg & Prof. Dr. Josef Brüderl

ECRs: Laura Schächtele & Daniel Krähmer

[Video-Introduction](#)

META-REP

Second funding phase (2025-2027)

Project 24: *SYNTH: Evaluating Transformer Models to Generate Synthetic Replications, Nomological Nets, and Peer Reviews*

PIs: Dr. Björn Hommel & Prof. Dr. Ruben Lennartz

ECRs: Beth Clarke, Annika Külpmann

Project 25: *Systematising Theory Building*

PIs: Dr. Xenia Schmalz

ECRs: Tatiana Logvinenko

[Video-Introduction](#)

[Video-Introduction](#)

META-REP

Second funding phase (2025-2027)

Project 26: *Theory Specification and Replicability*

PIs: Prof. Dr. Andreas Glöckner

ECRs: Fiona tho Pesch

[Video-Introduction](#)

Project 27: *From “Academic works” to “Working in academia”: Analyzing reform proposals in an agent-based modelling approach*

PIs: Prof. Dr. Felix Schönbrodt

ECRs: Kathie Harper

[Video-Introduction](#)

META-REP

Second funding phase (2025-2027)

Project 28: *The role of measurement in the replicability of empirical findings*

PIs: Dr. Susanne Frick & Prof. Dr. Eunike Wetzel

ECRs: Caroline Böhm & Steven Bißantz

[Video-Introduction](#)

Some notable tools coming from META-REP

- The **MetaPipeX Shiny App**: A tool to harmonize, analyze, visualize and document multi-lab data.
- The **SMART Shiny app**: A tool to guide researchers through and systematic definition and transparent documentation of multiverse analyses.
- The **g-fMRI-METEOR app** and The mEEG-METEOR app: Tools providing knowledge spaces of data processing and analysis decisions.
- For a collection of publications see:
<https://www.lmu.de/psy/de/forschung/meta-rep/results-publications/>

META-REP conference 2027

The 2027 META-REP Conference will take place from
November 30 to December 3, 2027
at the Katholische Akademie in Munich.

References A-G

- Abele-Brehm, A. E., Gollwitzer, M., Steinberg, U., & Schönbrodt, F. D. (2019). Attitudes Toward Open Science and Public Data Sharing: A Survey Among Members of the German Psychological Society. *Social Psychology*, 50(4), 252–260. <https://doi.org/10.1027/1864-9335/a000384>
- Bakker, M., Van Dijk, A., & Wicherts, J. M. (2012). The Rules of the Game Called Psychological Science. *Perspectives on Psychological Science*, 7(6), 543–554. <https://doi.org/10.1177/1745691612459060>
- Bargh, J. A., Chen, M., & Burrows, L. (o. J.). *Automaticity of Social Behavior: Direct Effects of Trait Construct and Stereotype Activation on Action*.
- Bem, D. J. (2011). Feeling the future: Experimental evidence for anomalous retroactive influences on cognition and affect. *Journal of Personality and Social Psychology*, 100(3), 407–425. <https://doi.org/10.1037/a0021524>
- Coles, N. A., March, D. S., Marmolejo-Ramos, F., Larsen, J. T., Arinze, N. C., Ndukaihe, I. L. G., Willis, M. L., Foroni, F., Reggev, N., Mokady, A., Forscher, P. S., Hunter, J. F., Kaminski, G., Yüvrük, E., Kapucu, A., Nagy, T., Hajdu, N., Tejada, J., Freitag, R. M. K., ... Liuzza, M. T. (2022). A multi-lab test of the facial feedback hypothesis by the Many Smiles Collaboration. *Nature Human Behaviour*, 6(12), 1731–1742. <https://doi.org/10.1038/s41562-022-01458-9>
- Doyen, S., Klein, O., Pichon, C.-L., & Cleeremans, A. (2012). Behavioral Priming: It's All in the Mind, but Whose Mind? *PLoS ONE*, 7(1), e29081. <https://doi.org/10.1371/journal.pone.0029081>
- Ebersole, C. R., Atherton, O. E., Belanger, A. L., Skulborstad, H. M., Allen, J. M., Banks, J. B., Baranski, E., Bernstein, M. J., Bonfiglio, D. B. V., Boucher, L., Brown, E. R., Budiman, N. I., Cairo, A. H., Capaldi, C. A., Chartier, C. R., Chung, J. M., Cicero, D. C., Coleman, J. A., Conway, J. G., ... Nosek, B. A. (2016). Many Labs 3: Evaluating participant pool quality across the academic semester via replication. *Journal of Experimental Social Psychology, Special Issue: Confirmatory*, 67, 68–82. <https://doi.org/10.1016/j.jesp.2015.10.012>
- Fiedler, K., Kutzner, F., & Krueger, J. I. (2012). The Long Way From α -Error Control to Validity Proper: Problems With a Short-Sighted False-Positive Debate. *Perspectives on Psychological Science*, 7(6), 661–669. <https://doi.org/10.1177/1745691612462587>
- Galak, J., LeBoeuf, R. A., Nelson, L. D., & Simmons, J. P. (2012). Correcting the past: Failures to replicate psi. *Journal of Personality and Social Psychology*, 103(6), 933–948. <https://doi.org/10.1037/a0029709>
- Gollwitzer, M., & Schwabe, J. (2022). Context Dependency as a Predictor of Replicability. *Review of General Psychology*, 26(2), 241–249. <https://doi.org/10.1177/10892680211015635>
- Gordon, M., Viganola, D., Bishop, M., Chen, Y., Dreber, A., Goldfedder, B., Holzmeister, F., Johannesson, M., Liu, Y., Twardy, C., Wang, J., & Pfeiffer, T. (2020). Are replication rates the same across academic fields? Community forecasts from the DARPA SCORE programme. *Royal Society Open Science*, 7(7), 200566. <https://doi.org/10.1098/rsos.200566>

References H-L

- Hardwicke, T. E., Thibault, R. T., Clarke, B., Moodie, N., Crüwell, S., Schiavone, S. R., Handcock, S. A., Nghiem, K. A., Mody, F., Eerola, T., & Vazire, S. (2024). Prevalence of Transparent Research Practices in Psychology: A Cross-Sectional Study of Empirical Articles Published in 2022. *Advances in Methods and Practices in Psychological Science*, 7(4), 25152459241283477. <https://doi.org/10.1177/25152459241283477>
- Holzmeister, F., Johannesson, M., Böhm, R., Dreber, A., Huber, J., & Kirchler, M. (2024). Heterogeneity in effect size estimates. *Proceedings of the National Academy of Sciences*, 121(32), e2403490121. <https://doi.org/10.1073/pnas.2403490121>
- Inbar, Y. (2016). Association between contextual dependence and replicability in psychology may be spurious. *Proceedings of the National Academy of Sciences*, 113(34), E4933–E4934. <https://doi.org/10.1073/pnas.1608676113>
- Ioannidis, J. P. A. (2005). Why Most Published Research Findings Are False. *CHANCE*, 18(4), 40–47. <https://doi.org/10.1080/09332480.2005.10722754>
- John, L. K., Loewenstein, G., & Prelec, D. (2012). Measuring the Prevalence of Questionable Research Practices With Incentives for Truth Telling. *Psychological Science*, 23(5), 524–532. <https://doi.org/10.1177/0956797611430953>
- Klein, R. A., Cook, C. L., Ebersole, C. R., Vitiello, C., Nosek, B. A., Hilgard, J., Ahn, P. H., Brady, A. J., Chartier, C. R., Christopherson, C. D., Clay, S., Collisson, B., Crawford, J. T., Cromar, R., Gardiner, G., Gosnell, C. L., Grahe, J., Hall, C., Howard, I., ... Ratliff, K. A. (2022). Many Labs 4: Failure to Replicate Mortality Salience Effect With and Without Original Author Involvement. *Collabra: Psychology*, 8(1), 35271. <https://doi.org/10.1525/collabra.35271>
- Klein, R. A., Ratliff, K. A., Vianello, M., Adams, R. B., Bahník, Š., Bernstein, M. J., Bocian, K., Brandt, M. J., Brooks, B., Brumbaugh, C. C., Cemalcilar, Z., Chandler, J., Cheong, W., Davis, W. E., Devos, T., Eisner, M., Frankowska, N., Furrow, D., Galliani, E. M., ... Nosek, B. A. (2014). Investigating Variation in Replicability: A “Many Labs” Replication Project. *Social Psychology*, 45(3), 142–152. <https://doi.org/10.1027/1864-9335/a000178>
- Klein, R. A., Vianello, M., Hasselman, F., Adams, B. G., Adams, R. B., Alper, S., Aveyard, M., Axt, J. R., Babalola, M. T., Bahník, Š., Batra, R., Berkics, M., Bernstein, M. J., Berry, D. R., Bialobrzeska, O., Binan, E. D., Bocian, K., Brandt, M. J., Busching, R., ... Nosek, B. A. (2018). Many Labs 2: Investigating Variation in Replicability Across Samples and Settings. *Advances in Methods and Practices in Psychological Science*, 1(4), 443–490. <https://doi.org/10.1177/2515245918810225>
- LeBel, E. P., McCarthy, R. J., Earp, B. D., Elson, M., & Vanpaemel, W. (2018). A Unified Framework to Quantify the Credibility of Scientific Findings. *Advances in Methods and Practices in Psychological Science*, 1(3), 389–402. <https://doi.org/10.1177/2515245918787489>
- Lewandowsky, S., & Oberauer, K. (2020). Low replicability can support robust and efficient science. *Nature Communications*, 11(1), 358. <https://doi.org/10.1038/s41467-019-14203-0>

References M-R

- *Many Labs 5: Testing Pre-Data-Collection Peer Review as an Intervention to Increase Replicability*. (o. J.). <https://doi.org/10.1177/2515245920958687>
- Marsh, A. A., Rhoads, S. A., & Ryan, R. M. (2019). A multi-semester classroom demonstration yields evidence in support of the facial feedback effect. *Emotion*, 19(8), 1500–1504. <https://doi.org/10.1037/emo0000532>
- Miller, J., & Ulrich, R. (2022). Optimizing Research Output: How Can Psychological Research Methods Be Improved? *Annual Review of Psychology*, 73(1), 691–718. <https://doi.org/10.1146/annurev-psych-020821-094927>
- Noah, T., Schul, Y., & Mayo, R. (2018). When both the original study and its failed replication are correct: Feeling observed eliminates the facial-feedback effect. *Journal of Personality and Social Psychology*, 114(5), 657–664. <https://doi.org/10.1037/pspa0000121>
- Nosek, B. A., Hardwicke, T. E., Moshontz, H., Allard, A., Corker, K. S., Dreber, A., Fidler, F., Hilgard, J., Kline Struhl, M., Nuijten, M. B., Rohrer, J. M., Romero, F., Scheel, A. M., Scherer, L. D., Schönbrodt, F. D., & Vazire, S. (2022). Replicability, Robustness, and Reproducibility in Psychological Science. *Annual Review of Psychology*, 73(1), 719–748. <https://doi.org/10.1146/annurev-psych-020821-114157>
- Nosek, B. A., Spies, J. R., & Motyl, M. (2012). Scientific Utopia: II. Restructuring Incentives and Practices to Promote Truth Over Publishability. *Perspectives on Psychological Science*, 7(6), 615–631. <https://doi.org/10.1177/1745691612459058>
- Oberauer, K., & Lewandowsky, S. (2019). Addressing the theory crisis in psychology. *Psychonomic Bulletin & Review*, 26(5), 1596–1618. <https://doi.org/10.3758/s13423-019-01645-2>
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251), aac4716. <https://doi.org/10.1126/science.aac4716>
- Patil, P., Peng, R. D., & Leek, J. T. (2016). What Should Researchers Expect When They Replicate Studies? A Statistical View of Replicability in Psychological Science. *Perspectives on Psychological Science*, 11(4), 539–544. <https://doi.org/10.1177/1745691616646366>
- Rosenthal, R. (1979). The file drawer problem and tolerance for null results. *Psychological Bulletin*, 86(3), 638–641. <https://doi.org/10.1037/0033-2909.86.3.638>
- Rubin, M. (2026, April 14). Is a 55% Replication Rate Too Low, Too High, or Just Right? [Substack newsletter]. *Critical Metascience*. <https://markrubin.substack.com/p/is-a-55-replication-rate-too-low>

References S-Z

- Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant. *Psychological Science*, 22(11), 1359–1366. <https://doi.org/10.1177/0956797611417632>
- Smaldino, P. E., & McElreath, R. (2016). The natural selection of bad science. *Royal Society Open Science*, 3(9), 160384. <https://doi.org/10.1098/rsos.160384>
- Strack, F. (2016). Reflection on the Smiling Registered Replication Report. *Perspectives on Psychological Science*, 11(6), 929–930. <https://doi.org/10.1177/1745691616674460>
- Strack, F., Martin, L. L., & Stepper, S. (1988). Inhibiting and facilitating conditions of the human smile: A nonobtrusive test of the facial feedback hypothesis. *Journal of Personality and Social Psychology*, 54(5), 768–777. <https://doi.org/10.1037/0022-3514.54.5.768>
- Van Bavel, J. J., Mende-Siedlecki, P., Brady, W. J., & Reinero, D. A. (2016). Contextual sensitivity in scientific reproducibility. *Proceedings of the National Academy of Sciences*, 113(23), 6454–6459. <https://doi.org/10.1073/pnas.1521897113>
- Vazire, S. (2018). Implications of the Credibility Revolution for Productivity, Creativity, and Progress. *Perspectives on Psychological Science*, 13(4), 411–417. <https://doi.org/10.1177/1745691617751884>
- Verhagen, J., & Wagenmakers, E.-J. (2014). Bayesian tests to quantify the result of a replication attempt. *Journal of Experimental Psychology: General*, 143(4), 1457–1475. <https://doi.org/10.1037/a0036731>
- Wagenmakers, E.-J., Beek, T., Dijkhoff, L., Gronau, Q. F., Acosta, A., Adams, R. B., Albohn, D. N., Allard, E. S., Benning, S. D., Blouin-Hudon, E.-M., Bulnes, L. C., Caldwell, T. L., Calin-Jageman, R. J., Capaldi, C. A., Carfagno, N. S., Chasten, K. T., Cleeremans, A., Connell, L., DeCicco, J. M., ... Zwaan, R. A. (2016). Registered Replication Report: Strack, Martin, & Stepper (1988). *Perspectives on Psychological Science*, 11(6), 917–928. <https://doi.org/10.1177/1745691616674458>
- Wagenmakers, E.-J., Wetzels, R., Borsboom, D., & Van Der Maas, H. L. J. (2011). Why psychologists must change the way they analyze their data: The case of psi: Comment on Bem (2011). *Journal of Personality and Social Psychology*, 100(3), 426–432. <https://doi.org/10.1037/a0022790>

Workshop

We will continue after a short break.



Workshop



- Find yourself in groups of 3-5 people!
Indicate your group number using your hand!
- Get in the breakout room for you group!
- Work through the following tasks and come back in between to discuss:



Guess the Replication



Google Colab Notebook

- As a group, play **Guess the Replication** @t1p.de/guther
Try to collect as many points as possible! (10min)
- Work together on the Google Colab task to explore replication rates:
https://colab.research.google.com/drive/1NN9CSLOWltLm9E3ATNT_4TJyRZfhoEre?usp=sharing
If you don't want to use Google Colab (or you don't have a Google account), you can also download the code and use any R interpreter. (30min)
- Discuss the questions on the next slide to reflect how the topic of replicability connects to your own research (20min)

You can find all the
tasks and links on
github.com/Schiwo/Sharing

Questions:

- What are your personal feelings when thinking about the replication crisis?
- Would you describe your own line of research as primarily exploratory or primarily confirmatory? Why?
- What do you see as the main challenges for improving replicability in your specific research field?
- How do you decide whether a finding is reliable in a) your own work and b) the work of other researchers?
- Have you ever encountered a situation in which being less transparent about problems in your research would have been the easier option? How did you decide how to handle the situation?